

Statistical Modeling

David Puelz

The University of Austin



- Use supervised learning to build models for prediction and for understanding (causal inference)
- Understand the benefits and limitations of the models we build
- Given a new situation, select an appropriate model, carry it out, and effectively communicate the results
- This course focuses on practice and implementation. But ... we will pick up theoretical details when necessary.



- **Educator:** Teaching at UATX since 2024, UT Austin from 2020-2024, graduate student from 2013-2018. Previously at the University of Chicago Booth School of Business and Goldman Sachs.
- **Researcher:** I develop new statistical/machine learning methods for analyzing complex data, with applications in policy, economics, and the social sciences.
- **Human:** Grew up in Dallas. Husband and dad with 2 girls + 1 boy + 1 dog. I like working out, smoking meat, golf, and cycling.



Introduction to the course

Course logistics

Supervised learning

Building intuition: k-nearest neighbors

The bias-variance tradeoff

Answer two questions for me:

What is one thing you're *excited* about for this class?

What is one thing you're *nervous or concerned* about for this class?

Pass around your answers. Another student will read your responses, so everyone stays anonymous.

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- Access at uaustin.populiweb.com
- Access at <https://github.com/dpuelz/StatisticalModeling>
- Github pages will contain **schedule**, **homeworks**, **data**, **code**, ...
- **Github will be our main landing page.** Populi will be used for grades and administrative tasks.



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- Class time will be mostly lecture and discussion. We will also have “in-class reps” on Quiz reviews and exercises.

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- Concepts assessed will come directly from the homeworks



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- I expect professional writeups in a markdown format (Rmd) or (Jupyter)





LLMs can be useful for the homeworks and learning concepts. I encourage their use! And ... I will not police their use. I only ask that you cite when LLMs are used.

However, if overused & relied on too heavily, **you will not learn anything.**

The quizzes are designed from the homeworks and will assess how well you understand the concepts. **Through the quizzes, you will get feedback quickly on whether you've overused AI!** Use this feedback to optimize your learning with AI.



- The exam will be during the 11th week (final exam week). There are no make-ups or alternative dates, so please plan accordingly!
- It will be a free response, written exam taken with pen and paper (exactly like our quizzes)



Homework	20%
Exam	30%
Quizzes	50%

- We will use **R** for statistical analysis throughout the course
- You are definitely welcome to use **Python**, but the course materials will (mostly) be in **R**
- This is industrial-strength, state-of-the-art, and free software for statistical computing
- We will access R through the **RStudio** graphical interface; make sure both are installed on your laptop and bring it to every class





- My office hours
- Schedule an one-on-one time with me to chat (Zoom or in-person)

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- Simply put, the goal is to predict a target variable Y with input variables X
- In data mining terminology, this is known as supervised learning
- We observe training data: $\{(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)\}$



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Tangent—what do you think **unsupervised learning** is?



In general, a useful way to think about it is that Y and X are related in the following way:

$$Y_i = f(X_i) + \epsilon_i$$



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$$Y_i = f(X_i) + \epsilon_i$$

- $f(X_i)$ is the systematic part (the signal)
- ϵ_i is the random error (the noise)
- The main purpose is to *learn or estimate* $f(\cdot)$ from data



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Today we'll focus on regression, but the same principles apply to classification



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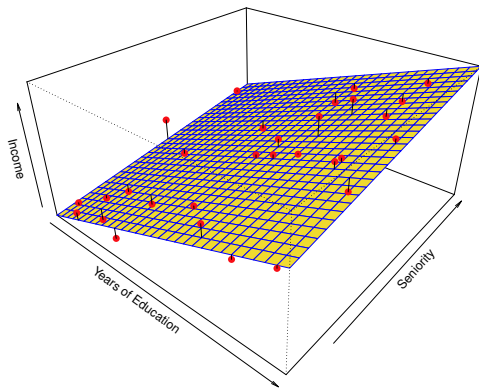
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 - **Parametric model**: Make assumptions about the form of f (e.g., linear)



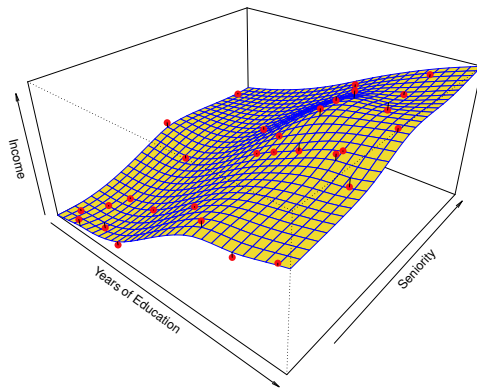
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3. Evaluate performance on **testing data** and *adjust*



Parametric



Nonparametric

- **Parametric:** Restrictive assumptions, simple interpretation
- **Nonparametric:** Flexible assumptions, complex interpretation

Introduction to the course

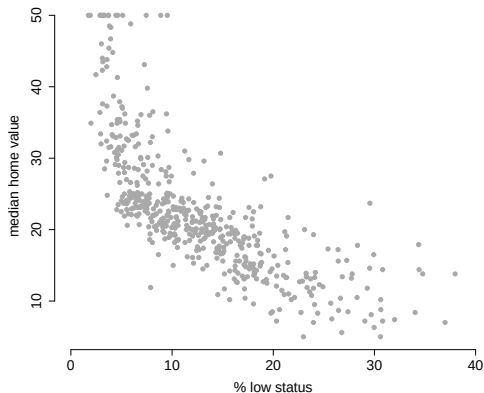
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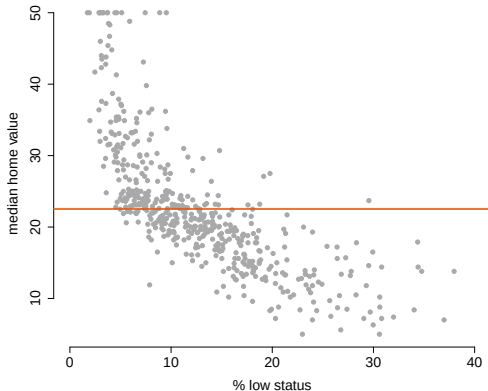
- Predict median home value with percent low economic status
- What should $f(\cdot)$ be?



What should $f(\cdot)$ be?

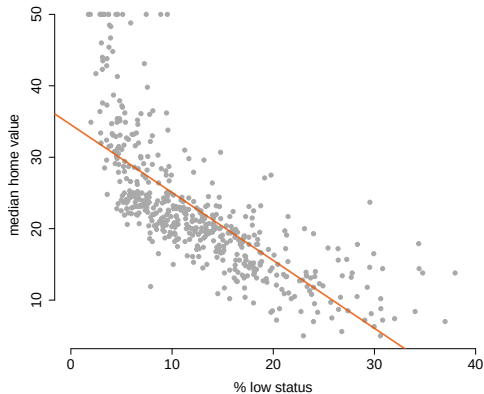


How about this? (Parametric: $Y = \mu + \epsilon$)



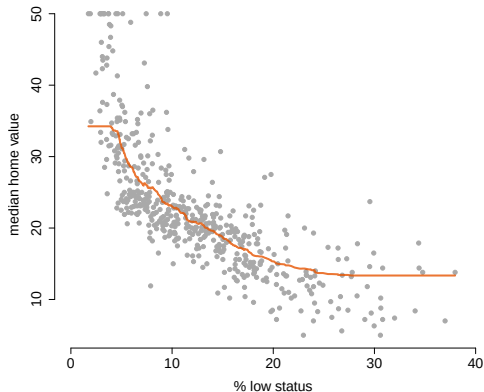
If % low status = 30, what is the prediction?

Linear model!



If % low status = 30, what is the prediction?

(k -NN with $k = 100$)



If % low status = 30, what is the prediction?



Prediction at point x : $\hat{f}(x)$ = average of k nearest points around x



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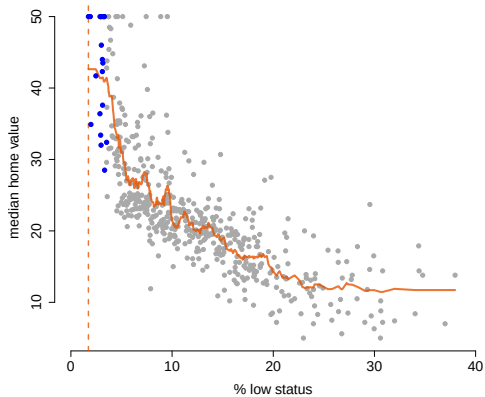
- Points that are **closer** to where we want to predict should be more relevant
- We choose the k points closest to the X value we're trying to predict
- This is the **k -nearest neighbors** algorithm

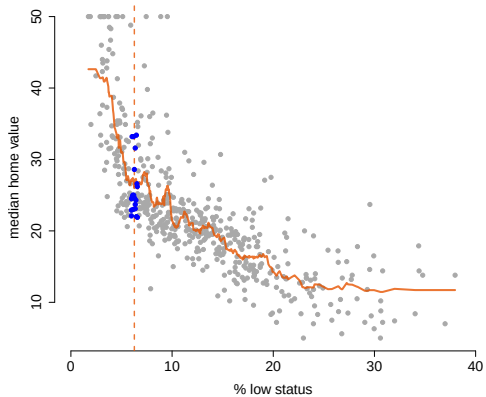


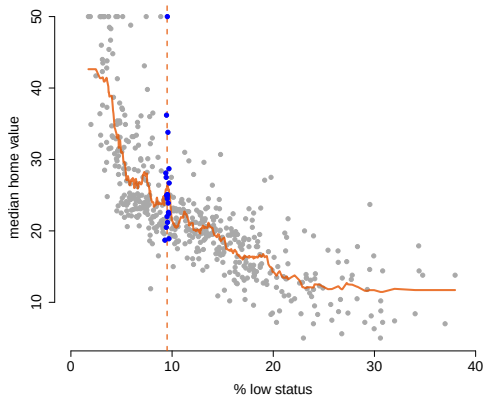
Prediction at point x : $\hat{f}(x)$ = average of k nearest points around x

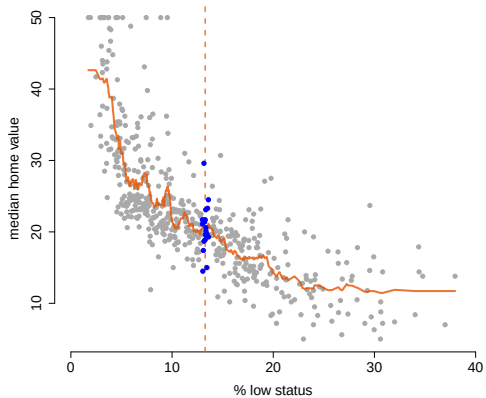
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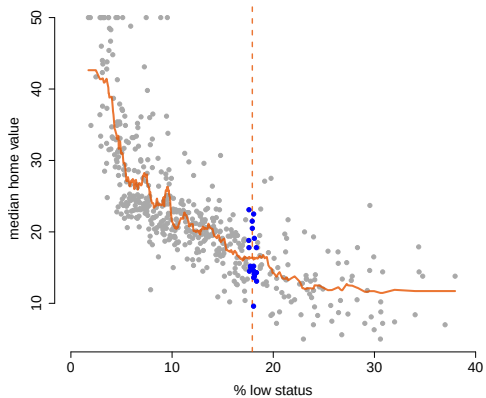
Let's see how this works with $k = 20$...

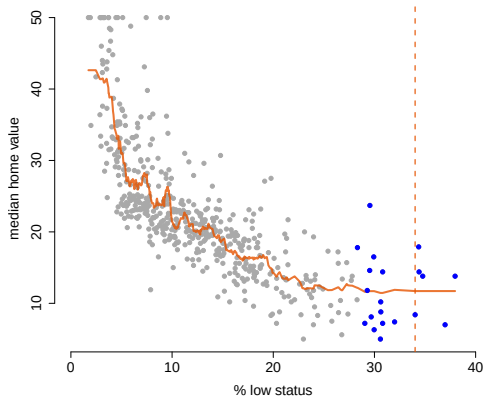




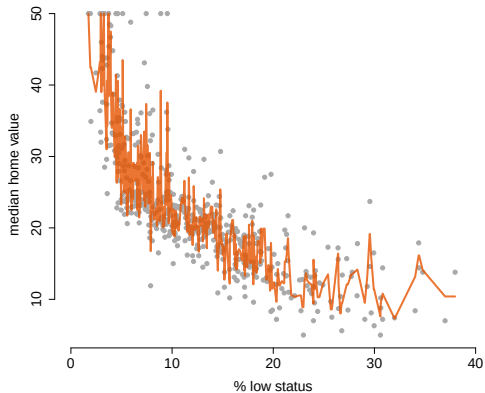




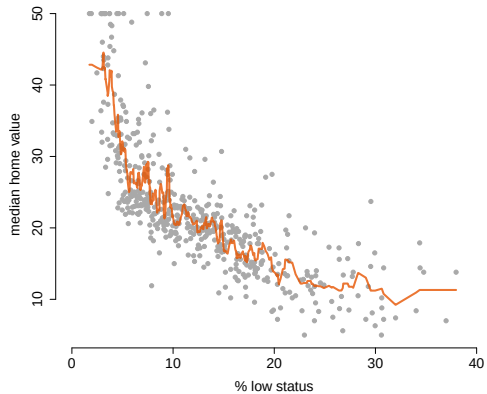




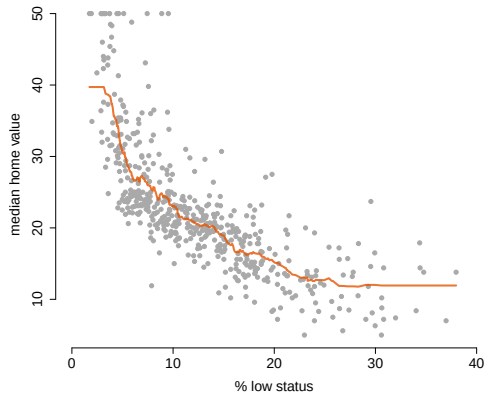
Why not choose $k = 2$?



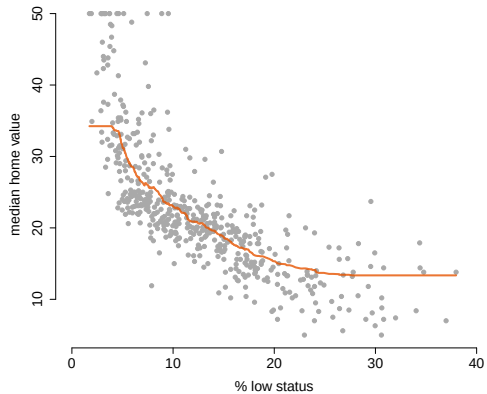
Or $k = 10$?



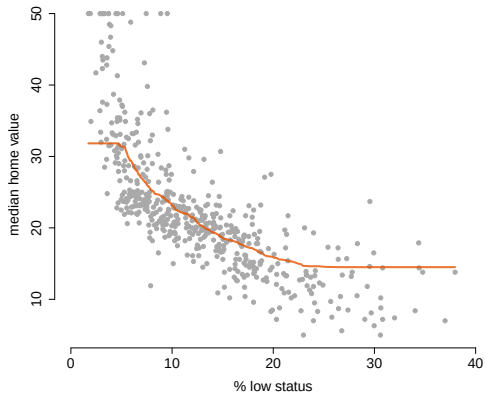
Or $k = 50$?



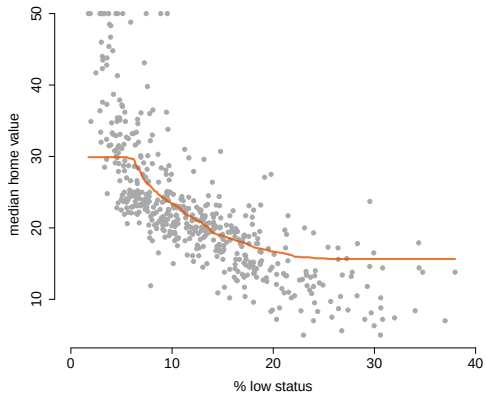
Or $k = 100$?



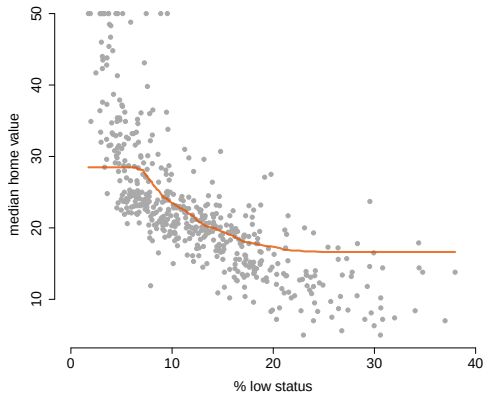
Or $k = 150$?



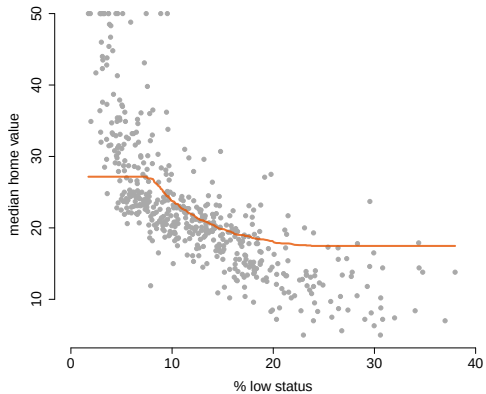
Or $k = 200$?



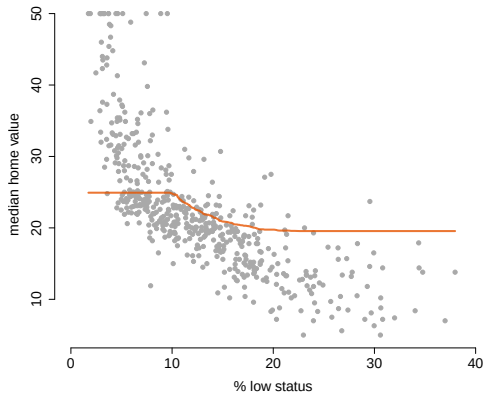
Or $k = 250$?



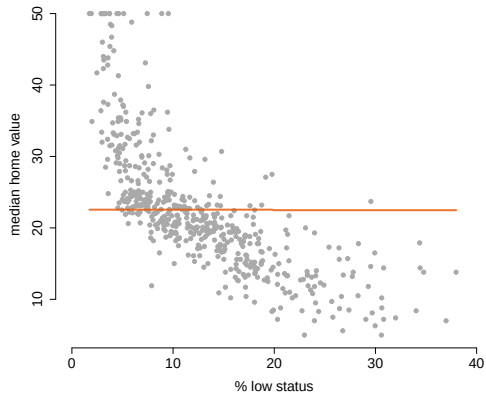
Or $k = 300$?



Or $k = 400$?



Or $k = 505$ (all data)?





- **In-sample error:** How well does the model fit the training data?

$$\text{RMSE}_{\text{in}} = \sqrt{\frac{1}{n} \sum_{i=1}^n [Y_i - \hat{f}(X_i)]^2}$$



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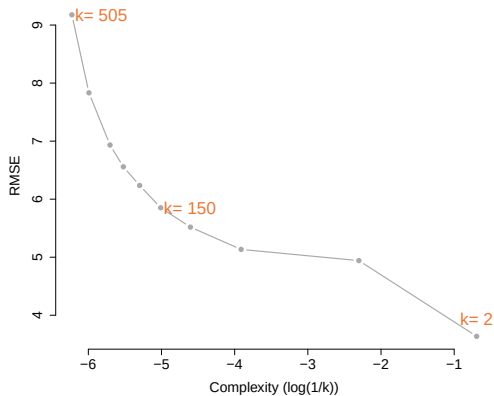
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- We care about out-of-sample performance!



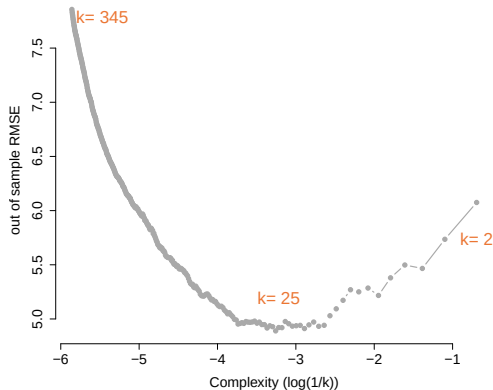
It looks like $k = 2$ is the best. Should we choose this model?



- Suppose we have m additional observations (X_i^o, Y_i^o) that we **did not use** to fit the model
- This is called the **validation set** (or hold-out set, or test set)
- We evaluate with out-of-sample RMSE:

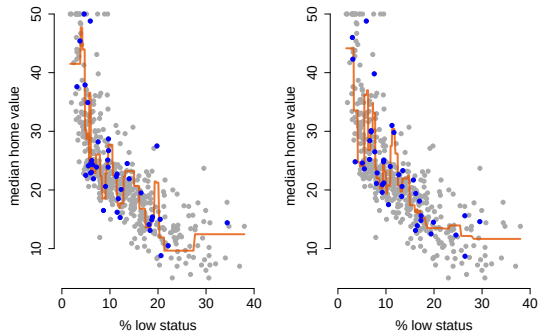
$$\text{RMSE}^o = \sqrt{\frac{1}{m} \sum_{i=1}^m [Y_i^o - \hat{f}(X_i^o)]^2}$$

Fit each model on training set (size 400). Test on validation set (size 106).



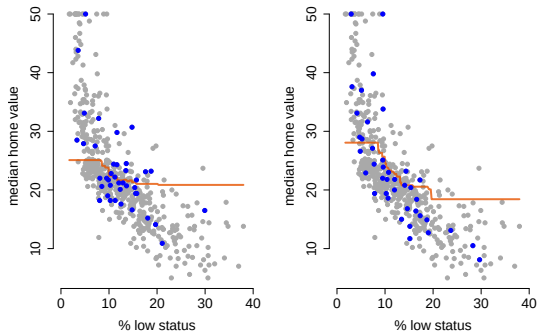
Now $k \approx 46$ looks like the best choice! Not too simple, but not too complex!

$k = 2$: Low bias, high variance



Training set of size 40

$k = 25$: High bias, low variance



Training set of size 40

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Our goal: Find the sweet spot in the bias-variance tradeoff!



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 - May not capture all patterns → **high bias**



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 - But predictions vary a lot → **high variance**



For a prediction at point x , the mean squared error can be decomposed:

$$\text{MSE} = \text{E}[(f(x) - \hat{f}(x))^2] = \underbrace{[\text{E}[\hat{f}(x)] - f(x)]^2}_{\text{Bias}^2} + \underbrace{\text{Var}[\hat{f}(x)]}_{\text{Variance}}$$



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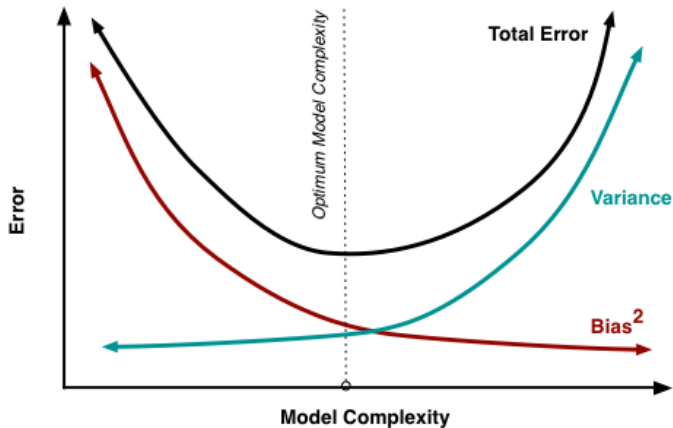
- **Bias²**: How far off our average prediction is from the truth



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- **Bias²**: How far off our average prediction is from the truth
- **Variance**: How much our predictions vary



- As model complexity increases, bias decreases but variance increases
- The optimal model complexity minimizes the total error



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 - Don't chase noise in the training data
- **The bias-variance tradeoff helps us understand this balance**



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4. We want to find the **sweet spot** that minimizes total prediction error



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 - Complex models: low bias, high variance
4. We want to find the **sweet spot** that minimizes total prediction error
5. **Out-of-sample performance** is what matters, not in-sample fit



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- The bias-variance tradeoff will continue to guide our thinking throughout the course!

Extra Slides



The prediction error $Y - \hat{f}(X)$ has three sources:

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Start with $Y = f(X) + \epsilon$. Decompose the error by adding/subtracting $E[\hat{f}(X)]$:

$$Y - \hat{f}(X) = [f(X) - E[\hat{f}(X)]] + [E[\hat{f}(X)] - \hat{f}(X)] + \epsilon$$



The prediction error $Y - \hat{f}(X)$ has three sources:

1. **Bias:** Our model's average prediction $E[\hat{f}(X)]$ might be off from the true $f(X)$
2. **Variance:** Our specific prediction $\hat{f}(X)$ varies around its average
3. **Noise:** The true Y has random error ϵ even if we knew $f(X)$ perfectly

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Squaring both sides gives squared terms **plus cross terms**:

$$\begin{aligned}(Y - \hat{f}(X))^2 &= [f(X) - E[\hat{f}(X)]]^2 + [E[\hat{f}(X)] - \hat{f}(X)]^2 + \epsilon^2 \\ &\quad + (\text{cross terms})\end{aligned}$$



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For the MSE $E[(f(X) - \hat{f}(X))^2]$, the cross terms vanish, leaving only the two squared terms: bias^2 and variance.